

# Python CSV File Handling

Reference Solutions (Menu Driven Programs)

## Program 1: Student Record Management System

```
import csv

FILE = "students.csv"

def add_student():
    with open(FILE, "a", newline="") as f:
        writer = csv.writer(f)
        roll = input("Roll No: ")
        name = input("Name: ")
        course = input("Course: ")
        marks = input("Marks: ")
        writer.writerow([roll, name, course, marks])

def view_students():
    with open(FILE, "r") as f:
        reader = csv.reader(f)
        for row in reader:
            print(row)

while True:
    print("\n1.Add Student  2.View Students  3.Exit")
    ch = input("Enter choice: ")
    if ch == "1":
        add_student()
    elif ch == "2":
        view_students()
    elif ch == "3":
        break
```

## Program 2: Employee Salary Register

```
import csv

FILE = "employees.csv"

def add_employee():
    with open(FILE, "a", newline="") as f:
        writer = csv.writer(f)
        eid = input("Emp ID: ")
        name = input("Name: ")
        dept = input("Department: ")
        salary = input("Salary: ")
        writer.writerow([eid, name, dept, salary])

def search_employee():
    emp_id = input("Enter Employee ID to search: ")
    with open(FILE, "r") as f:
        reader = csv.reader(f)
        for row in reader:
            if row and row[0] == emp_id:
                print(row)
                return
    print("Employee not found")

while True:
    print("\n1.Add    2.Display All    3.Search    4.Exit")
    ch = input("Choice: ")
    if ch == "1":
        add_employee()
    elif ch == "2":
        with open(FILE, "r") as f:
            for r in csv.reader(f):
                print(r)
    elif ch == "3":
        search_employee()
    elif ch == "4":
        break
```

## Program 3: Product Inventory System

```
import csv

FILE = "products.csv"

def add_product():
    with open(FILE, "a", newline="") as f:
        writer = csv.writer(f)
        pid = input("Product ID: ")
        name = input("Product Name: ")
        price = input("Price: ")
        qty = input("Quantity: ")
        writer.writerow([pid, name, price, qty])

def low_stock():
    limit = int(input("Enter quantity limit: "))
    with open(FILE, "r") as f:
        reader = csv.reader(f)
        for row in reader:
            if row and int(row[3]) < limit:
                print(row)

while True:
    print("\n1.Add 2.View 3.Low Stock 4.Exit")
    ch = input("Choice: ")
    if ch == "1":
        add_product()
    elif ch == "2":
        with open(FILE, "r") as f:
            for r in csv.reader(f):
                print(r)
    elif ch == "3":
        low_stock()
    elif ch == "4":
        break
```

## Program 4: Attendance Management System

```
import csv

FILE = "attendance.csv"

def mark_attendance():
    with open(FILE, "a", newline="") as f:
        writer = csv.writer(f)
        date = input("Date: ")
        roll = input("Roll No: ")
        name = input("Name: ")
        status = input("Status (P/A): ")
        writer.writerow([date, roll, name, status])

def view_by_date():
    date = input("Enter date: ")
    with open(FILE, "r") as f:
        for row in csv.reader(f):
            if row and row[0] == date:
                print(row)

while True:
    print("\n1.Mark 2.View by Date 3.View All 4.Exit")
    ch = input("Choice: ")
    if ch == "1":
        mark_attendance()
    elif ch == "2":
        view_by_date()
    elif ch == "3":
        with open(FILE, "r") as f:
            for r in csv.reader(f):
                print(r)
    elif ch == "4":
        break
```

## Program 5: Library Management (Edit & Delete)

```
import csv

FILE = "library.csv"

def read_all():
    with open(FILE, "r") as f:
        return list(csv.reader(f))

def write_all(data):
    with open(FILE, "w", newline="") as f:
        csv.writer(f).writerows(data)

def update_status():
    bid = input("Book ID: ")
    data = read_all()
    for row in data:
        if row and row[0] == bid:
            row[3] = input("New Status: ")
    write_all(data)

def delete_book():
    bid = input("Book ID to delete: ")
    data = read_all()
    data = [row for row in data if row and row[0] != bid]
    write_all(data)

while True:
    print("\n1.Add 2.View 3.Update 4.Delete 5.Exit")
    ch = input("Choice: ")
    if ch == "1":
        with open(FILE, "a", newline="") as f:
            csv.writer(f).writerow([
                input("ID: "), input("Name: "),
                input("Author: "), input("Status: ")
            ])
    elif ch == "2":
        for r in read_all():
            print(r)
    elif ch == "3":
```

```
        update_status()
elif ch == "4":
    delete_book()
elif ch == "5":
    break
```

**End of Reference Solutions**